Enrollment No.: 240840116014

Practical 8

Create interactive visualizations using Plotly Express and Plotly Graph Objects.

Step 1: Import Required Libraries

```
# Import Pandas.
import pandas as pd

# Import Plotly Express.
import plotly.express as px

# Import Ploty Graph Objects.
import plotly.graph_objects as go
```

Step 2: Create Dataset

```
data = {
    "Month": ["January", "February", "March", "April", "May", "June", "July"
    "Product": ["Laptop", "Mobile", "Tablet", "Laptop", "Mobile", "Tablet",
    "Region": ["North", "South", "East", "West", "North", "South", "East", "
    "Sales": [120000, 135000, 150000, 145000, 160000, 175000, 180000, 170000
    "Units_Sold": [12, 30, 25, 14, 35, 30, 18, 38, 32, 21, 45, 40],
    "Profit": [18000, 27000, 30000, 22000, 32000, 35000, 28000, 34000, 38000
}
df = pd.DataFrame(data)
df
```

	Month	Product	Region	Sales	Units_Sold	Profit
0	January	Laptop	North	120000	12	18000
1	February	Mobile	South	135000	30	27000
2	March	Tablet	East	150000	25	30000
3	April	Laptop	West	145000	14	22000
4	May	Mobile	North	160000	35	32000
5	June	Tablet	South	175000	30	35000
6	July	Laptop	East	180000	18	28000
7	August	Mobile	West	170000	38	34000
8	September	Tablet	North	190000	32	38000
9	October	Laptop	South	210000	21	33000
10	November	Mobile	East	2300010	45	46000
11	December	Tablet	West	250000	40	50000

Step 3: Inspect the Dataset

```
# Display First Five Columns.
df.head()
```

	Month	Product	Region	Sales	Units_Sold	Profit
0	January	Laptop	North	120000	12	18000
1	February	Mobile	South	135000	30	27000
2	March	Tablet	East	150000	25	30000
3	April	Laptop	West	145000	14	22000
4	May	Mobile	North	160000	35	32000

```
# Display No. of Rows and Columns.
df.shape
```

```
(12, 6)
```

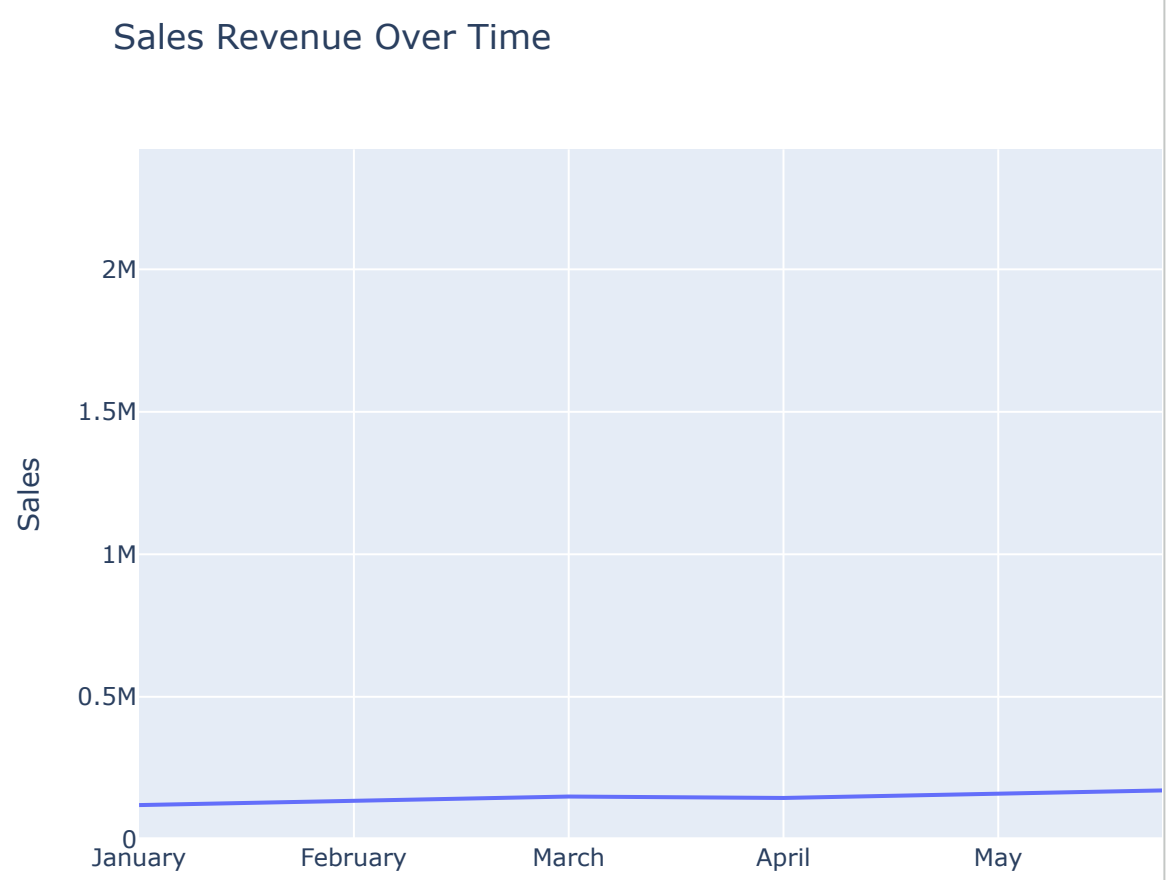
```
# Display Information od dataset.
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 12 entries, 0 to 11
Data columns (total 6 columns):
 #   Column      Non-Null Count  Dtype 
---  ------      --------------  ----- 
 0   Month       12 non-null     object
```

```
 1   Product     12 non-null     object
 2   Region      12 non-null     object
 3   Sales       12 non-null     int64
 4   Units_Sold  12 non-null     int64
 5   Profit      12 non-null     int64
dtypes: int64(3), object(3)
memory usage: 708.0+ bytes
```

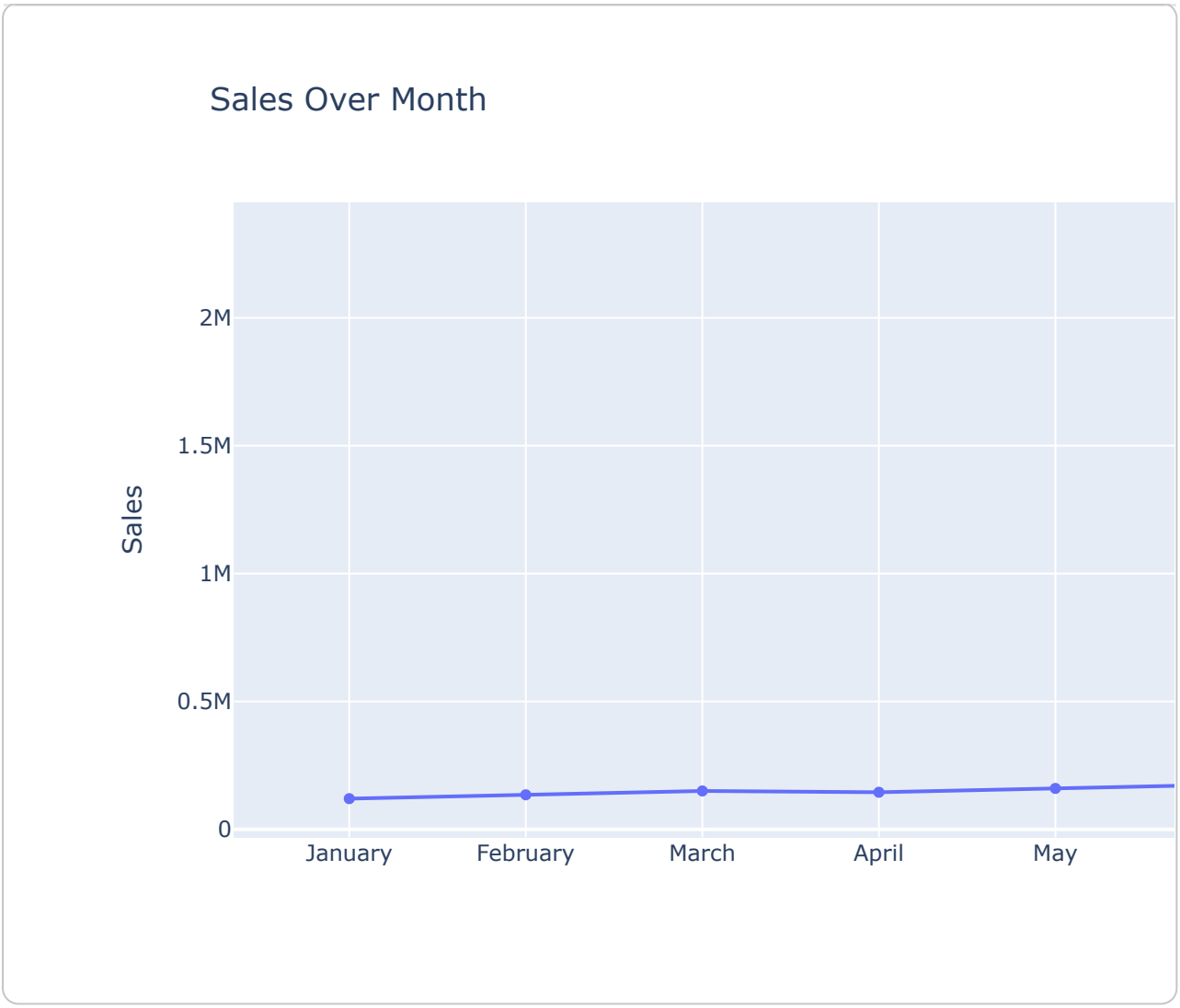
Step 4: Create an Interactive Line chart using Plotly Explore

```
# How does the sales revenue change throughout the year?
fig = px.line(df, x="Month", y="Sales", title="Sales Revenue Over Time")
fig.show()
```

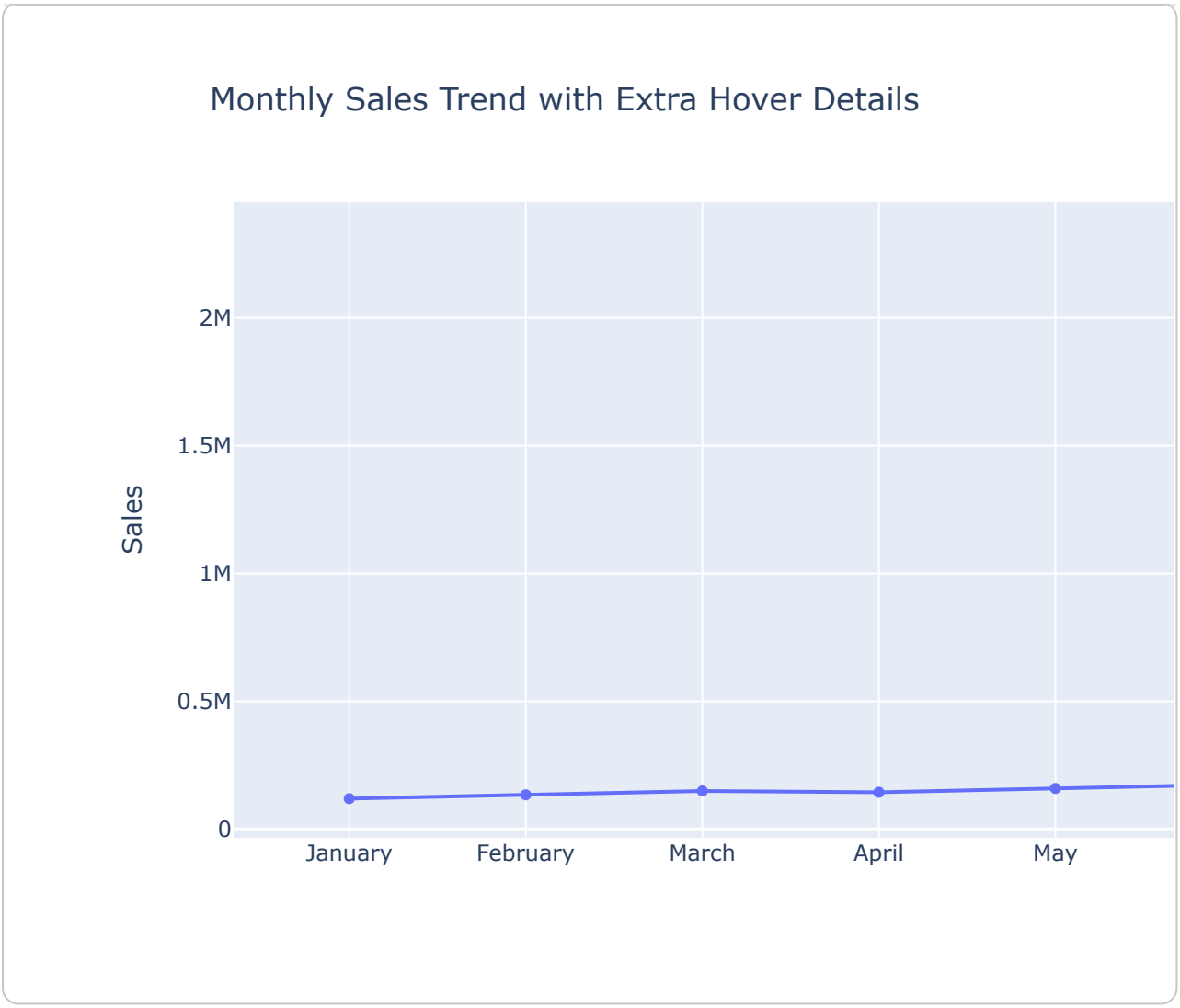


Step 5: Improve the Hover Information

```
# Create a Line Chart using a Sales over Month.
fig = px.line(df, x='Month', y='Sales', title='Sales Over Month', markers=Tr
fig.show()
```




```
# Add additional columns to the hover information. Student can see Product,
import plotly.express as px
fig = px.line(df,x="Month",y="Sales",markers=True,title="Monthly Sales Trend
fig.show()
```



Step 6: Create an Interective Bar Chart

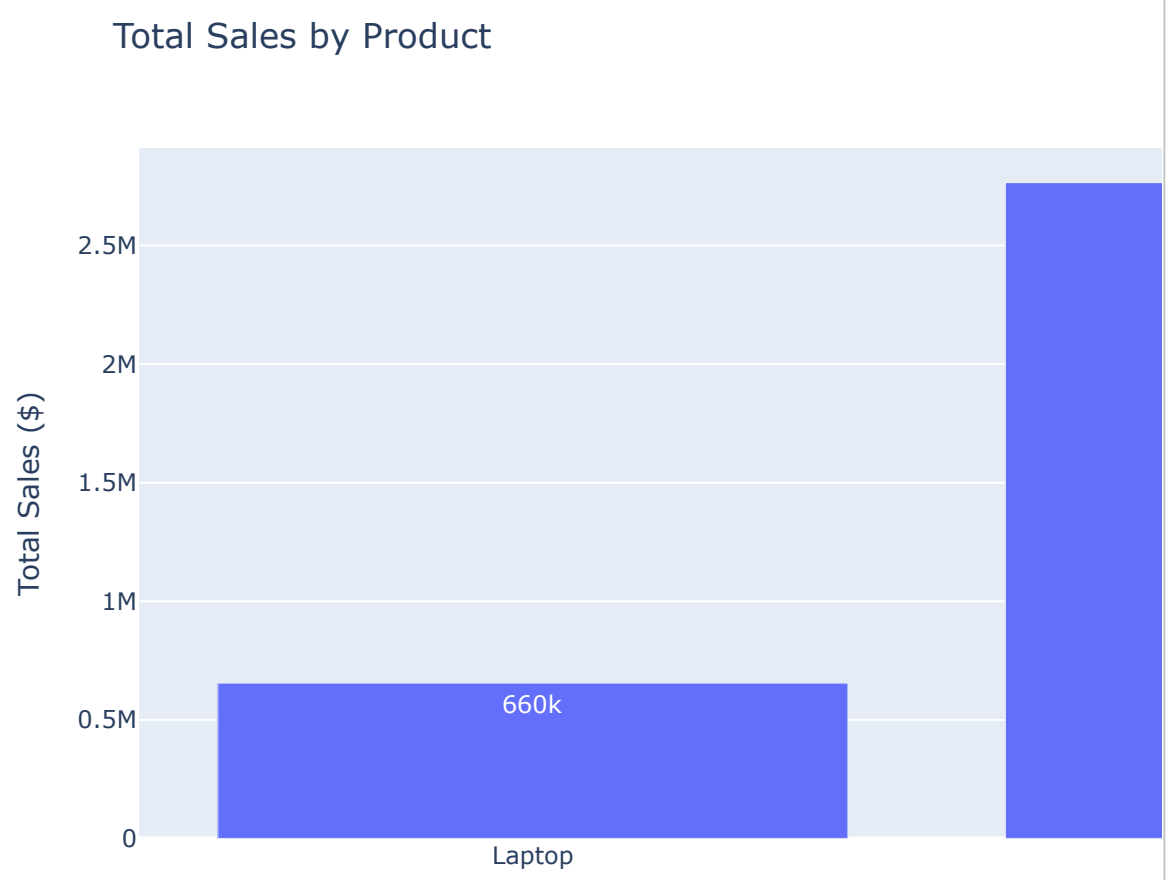
```
# Group the datset accordimg to Product.
product_sales = df.groupby('Product')['Sales'].sum().reset_index()
```

```
# Calculate the total Sales generated by each product.
df.groupby('Product')['Sales'].sum().reset_index()
```

	Product	Sales
0	Laptop	655000
1	Mobile	2765010
2	Tablet	765000

```
# Create an interective bar chart.
fig = px.bar(
    product_sales,
    x='Product',
    y='Sales',
    title='Total Sales by Product',
    labels={'Sales': 'Total Sales ($)', 'Product': 'Product Name'},
```

```
    text_auto='.2s'
)
fig.show()
```



Step 7: Interective Scatter Plot

```
# TO analyze the relationship between Units Sold and Sales Using an Interect
fig = px.scatter(
    df,
    x="Units_Sold",
    y="Sales",
    title="Units Sold vs Sales",
    labels={"Units_Sold": "Units Sold", "Sales": "Sales ($)"}
)
fig.show()
```



Step 8: Interective Scatter Plot with Color

```
# Create an interactive scatter plot with color
fig = px.scatter(
    df,
    x="Units_Sold",
    y="Sales",
    color="Product",
    title="Units Sold vs Sales by Product",
    labels={"Units_Sold": "Units Sold", "Sales": "Sales ($)"}
)
fig.show()
```

